

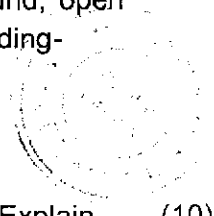
Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 1st Term Regular Examination, 2023
URP 4171
(Legal Basis of Planning)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

- 1 (a) What is your understanding about the legal aspects of planning? (05)
(b) Narrate the factors on the basis of which compensation for the acquisition of land is determined. (15)
(c) Explain how the involvement of different statutes leads to regulatory intervention in the land acquisition process in Bangladesh, providing necessary examples. (15)
- 2 (a) Mention the maps and clearance certificates that have to be submitted for approving a private housing project. (12)
(b) Suppose you are working as an urban planner in a real estate company and you have to prepare a layout plan for a private housing project. Make a list of the documents you require to be registered from the concerned authority. (08)
(c) Briefly discuss the significant amendments that were made in the "Private Housing Project Land Development Rule, 2004" to establish the "Private Housing Project Land Development Rule, 2012." (15)
- 3 (a) Critically analyze the shortcomings of "Mega city, Divisional Town and District Town's municipal areas including country's all the municipal areas' playground, open space, park and natural water reservoir Conservation Act, 2000". (08)
(b) "According to the New York City Building Zone Resolution, 1916, New York city has been divided as Use Districts, Height Districts & Area Districts." Briefly explain the districts. (12)
(c) What does the Secretary of State need to consider in reaching a decision on the designation of a new town? Briefly explain. (15)
- 4 (a) Make a brief overview of the process of designating a new town according to "New Towns Act 1981 (Local Authority Oversight) Regulations 2018" (15)
(b) Briefly discuss the direction of "Mega city, Divisional Town and District Town's municipal areas including country's all the municipal areas' playground, open space, park and natural water reservoir Conservation Act, 2000" regarding- (10)
 - i. Offences committed by companies;
 - ii. Penalties
(c) How the planning regulations of Japan are different from our country? Explain. (10)

Section – B

- 5 (a) What do you mean by National Building Code (NBC)? (05)
(b) Briefly discuss some important building codes and documents that were reviewed for the formulation of BNBC. (10)

- (c) Critically analyze the planning aspects that have been considered in the updated BNBC. (20)
- 6 (a) Discuss the governance and functions of DIT Trustee Board. (10)
- (b) Analyze the causes and consequences of deviation of building constructions in the urban areas of Bangladesh. (15)
- (c) Discuss the functions of RAJUK. (10)
- 7 (a) Elaborate the directives on “Master Plan” and “Site Development Schemes” under the Heading of Town Planning of the Local Government (Municipalities) Act 2009. (15)
- (b) Write notes on the followings: (20)
- i. Declaration of city area and establishment of municipality;
 - ii. Property and management of funds for the municipality.
- 8 (a) Formulation and execution of URP Act 2017 requires coordinated efforts of many interrelated ministries and departments of Bangladesh. Justify this statement. (15)
- (b) Narrate the functions of the Urban and Regional Planning Advisory Council under the draft URP Act 2017. (10)
- (c) Discuss the mandates and functions of UDD under the URP Act 2017. (10)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 1st Term Examination, 2023.
URP 4141
(Environmental Planning and Management)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script
ii) Formula chart should be provided.
iii) Assume necessary data if required.
iv) Figures in the right margin indicate full marks.

Section – A

1. (a) Differentiate between energy flow and nutrient cycle in ecosystem with necessary diagram. (07)
- (b) Describe four ecosystem services according to millennium ecosystem assessment (MEA) of United Nation. (08)
- (c) The current albedo rate at top of atmosphere combined with the earth surface is approximately 0.29. If this were increased by 1.1%, what would be the amount of solar heat flux reaching on earth? Again there is currently about 1.29 W/m² heat surplus on earth. If ocean covers about 71% of the earth surface, how much excess heat in petawatts may be going into ocean and land surface. Consider 1PW = 10¹⁵ W and earth radius is 6378 km. (10)
- (d) The population of 'X' region is 2,70,000 and it produces 1,20,000 tons of rice annually. The global average yield of cropland is 1.15 tons/hector/year and the global average capability of cropland is 1.75 tons/hector/year. If 'X' region also imports 30,000 ha of grape and 15,000 ha of corn annually. What would be the ecological footprint of consumption in per capita for 'X' region for that year? (10)
2. (a) What do you mean by radiative forcing? Describe why greenhouse gases have positive (+ve) and aerosol has negative (-ve) radiative forcing value. (08)
- (b) Describe the 'Green House Effect' with a perspective of earth energy exchange. (07)
- (c) Temperature increase accelerates the air borne diseases. Consider a 100-120nm virus is travelling in air at temperature (T) 27°C. If the drag coefficient is 1.28×10⁻²² Kg/s how much linear distance it will travel after half an hour? If temperature increase in summer by 4% that makes the drag coefficient 1.28×10⁻²² Kg/s, how much distance the virus will travel then at the same time? (Boltzmann constant = 1.38 × 10⁻²³ J/k) (10)
- (d) Assume 1,50,000 people of 'Y' city on an average give a round trip to 'Z' city on a monthly basis. What will be the yearly passenger travel based carbon emission in tons for this route by bus? Assume:
 - EF for diesel bus = 0.5281 kgCO₂/vkm.
 - Distance between Y and Z = 159 miles.
3. (a) Describe the followings: (10)
 - i) Urban heat island ii) Why ozone is both good and bad?
- (b) Illustrate the concept of emission trading system (ETS) and carbon tax in terms of carbon pricing mechanism. (10)
- (c) Suppose that a certain company now believes that the probability of a favorable market exactly same as the probability of an unfavorable market. Thus, each state of nature has a 0.5 probability. (15)

Alternatives	State of Nature	
	Favorable market	Unfavorable market
Construct a large plant	1,80,000\$	-1,60,000\$
Construct a small plant	90,000\$	-15,000\$
Do nothing	0	0
Probability (0-1)	0.5	0.5



Which decision would be best under following scenario? Also, perform the sensitivity analysis and represent the range of a p values with alternatives.

4. (a) Describe the environmental externalities accelerated market failure with price quantity curve, showing the internalizing of it. (12)
- (b) Mention major limitations of command and control instruments (CAC) in environmental management. (08)
- (c) What are the market based instrument (MBI) in environmental regulation? Describe some MBIs with examples. (15)

Section – B

5. (a) Why EIA is essential for implementation of any large-scale project? Discuss with reference to negative impacts of development and importance of environment management. (15)
- (b) Write down the differences between EIA and SEA (08)
- (c) Discuss the nexus between Economic growth, Environmental Degradation and Poverty (12)
6. (a) Which package of impact identification and prediction method will you apply for both qualitative and quantitative data visualization (10)
- (b) As an environmental planner what will you consider while impact identification. (13)
- (c) Discuss the characteristics of environmental impacts. (12)
7. (a) If you are assigned to a project as an Environment Planner and you have to design mitigation measure for "Rupsa Textile Industry". The textile industry emitted toxic waste including caustic soda, sulphides, acetic acid, metallic salts, starch. Under some following condition design, a mitigation measure following "Mitigation Measure Hierarchy" (15)
 1. Area is a suburban area
 2. Industry is near settlement
 3. Beside a river
 4. Soil type of project site is less compact. x
- (b) "The developed countries have good living environment in exchange of the dirt they produce in underdeveloped and developed counties"- Justify the statement making conjunction between Kuznet curve and Pollution Haven Theory. (15)
- (c) Mention the alternative options need to be examined for projects. (05)
8. (a) What should a good Environment Management Plan include? (06)
- (b) "The more we develop, the more environment gets degraded" – Explain with Material Balance Model. (15)
- (c) Briefly discuss the major types of stakeholder's roles in different stages of EIA formation. (14)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 1st Term Regular Examination, 2023
URP 4161
(Project Planning and Management)

Full Marks: 210

Time: 3.00 hrs

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

- 1 (a) Illustrate-Vision, Mission, Policy, Strategy, Program and Project with necessary examples. (24)
- (b) Illustrate Project Life cycle with the help of a diagram. (11)
- 2 (a) What is Logical Framework Approach? (05)
- (b) Prepare a Log Frame as a part of Development Project Proposal for Urban Sanitation project. (15)
- (c) Discuss the various plans that are involved in the planning phase of project management cycle. (15)
- 3 (a) What do you mean by Result Based Monitoring? (05)
- (b) Discuss various types of monitoring with necessary example. (15)
- (c) Discuss various types of evaluation with necessary example. (15)
- 4 (a) Discuss on the phases involved in a project's stakeholder analysis. (15)
- (b) Draw your arguments in managing conflict in project management. (10)
- (c) Distinguish among public and private project in Bangladesh. (10)

Section – B

- 5 (a) "A taka today is more valuable than a taka a year hence"-justify the statement. (05)
- (b) What are the merits and demerits of different types of investment criteria? (10)
- (c) A firm borrows Rs. 500,000 at an interest rate of 15%, and the loan is to be repaid in 4 equal installments payable at the end of each of the next 5 years. Create a loan amortization schedule showing the breakdown of each payment into principal and interest, the remaining balance after each payment, and the total interest paid over these 5 years. (20)
- 6 (a) Calculate the Discounted Payback Period, where discount rate=10% (20)

Year	Cash Flow (\$)
0	10,00000
1	150000
2	200000
3	250000
4	300000
5	350000
6	400000

- (b) Distinguish between ordinary annuity and annuity due. (10)
- (c) What do you mean by break-even point? (05)

- 7 (a) Write short notes on the followings: i) Incremental principle, ii) Consistency principle. (10)

- (b) Find the operating cash flow from the following table: (15)

	Year 1-5
Project revenues (growth rate, 0%)	240000
Cost of goods sold (60% of revenues)	144,000
Cash operating expense	5,000
Depreciation	40,000
Taxes (30%)	15,300

- (c) Describe the time horizon of project cash flow. (10)

- 8 (a) The Apollo Group limited had revenue of Tk.150 million in 2000, which increased to tk.1500 million in 2010. What was the compound growth rate in revenue? (10)

- (b) Why does the discounted payback period get priority over payback period? (07)

- (c) Differentiate between simple and compound interest with necessary examples. (10)

- (d) Illustrate the pros and cons of Benefit Cost Ratio. (08)

Khulna University of Engineering & Technology
Department of Urban and Regional Planning
BURP 4th Year 1st Term Regular Examination, 2023
URP 4175
(Resettlement Planning)

Full Marks: 210

Time: 3.00 hrs.

- N.B.** i) Answer any three questions from each section in separate script.
ii) Figures in the right margin indicate full marks.

Section – A

- 1 (a) Provide a brief history of Rohingya displacement and discrimination in Myanmar/Bangladesh. (12)
(b) "Fire Hazards in Slums/Low-cost markets are the new form of involuntary/ voluntary displacement- To what extent do agree/disagree with the statement. Justify your answer with proper logic. (13)
(c) Write down the problems regarding to resettlement. (10)
- 2 (a) Define RAP. Mention the components of RAP. (10)
(b) Why do Legal framework and Compensation are given priority as first and second most important instruments of RAP? (10)
(c) Correlate Resettlement plan with project cycle steps. (15)
- 3 (a) Write down the social, political and economic problem for Bangladesh in the coming future? (15)
(b) Write short note on: i) Resettlement Plan Due ii) Community Compensation (10)
(c) Discuss about all types of eligibility criteria for compensation in RPF. (10)
- 4 (a) Briefly discuss the methods of compensation payments. (12)
(b) Write down the issues concerning with institutional framework. (13)
(c) Mention the roles can be played by affected persons in different stages of resettlement project. (10)

Section – B

- 5 (a) Should resettlers be involved in the resettlement decisions? Explain. (10)
(b) Address some of the loopholes for finalizing land price in the 1982 ARIPO Ordinance. (10)
(c) Provide recommendation for the following issues in the economics of resettlement: (15)
i) Avoiding involuntary resettlement
ii) Property Rights
iii) Full and proper compensation
- 6 (a) Elaborate on the risks to displaced person / communities after resettlement. (15)
(b) "The present resettlement policies are adequate" – Give your arguments for or against this statement. (15)
(c) What are the impacts covered by the World Bank on Involuntary Resettlement? (05)
- 7 (a) According to the IRR Model, mention some of the risks and their counterparts. (10)
(b) Explain the limitations of cost-benefit analysis for involuntary resettlement. (10)
(c) Describe the issues and difficulties in planning and implementation of resettlement economics. (15)
- 8 (a) What are the advantages and limitations of GRM? (10)
(b) What are the advantages of integrating IRR model with current resettlement practice? (10)
(c) Present a case study to describe how the lessons learnt from past resettlement projects are helpful. (15)



Khulna University of Engineering & Technology
 Department of Urban and Regional Planning
 BURP 4th Year 1st Term Regular Examination, 2003
URP 4111
 (Utility Services and Infrastructure Planning)

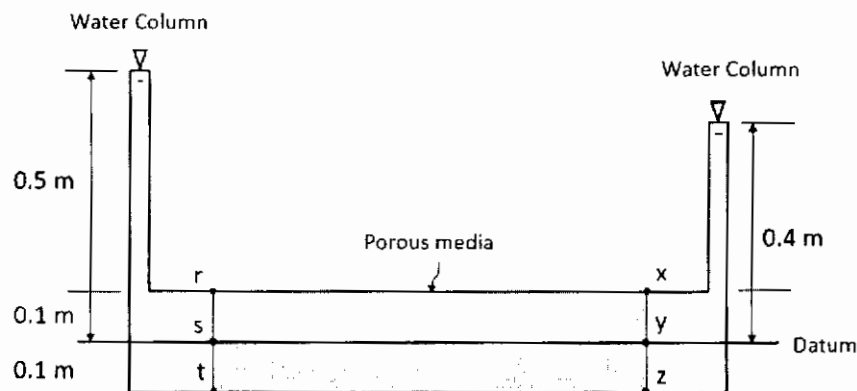
Full marks: 210

Time: 3.0 hours

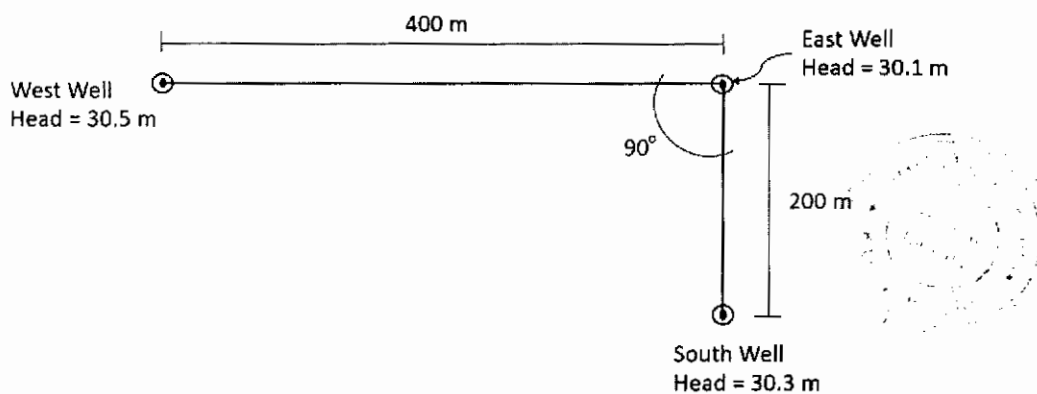
- N.B: (i) Answer any **three questions** from each section in separate script.
 (ii) Figure in the right margin indicates full marks.
 (iii) Use the formula chart if required.
 (iv) Assume necessary data if required.

Section- A

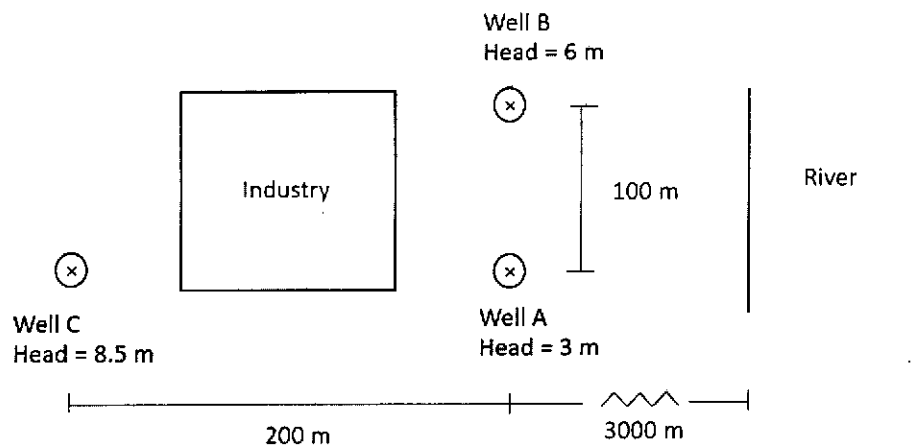
1. (a) Illustrate your concept about 'Unaccounted-for-water'. (05)
- (b) Calculate the value of total head (H_t) for each point in the equipotential lines of r,s,t and x,y,z from the following figure. (07)



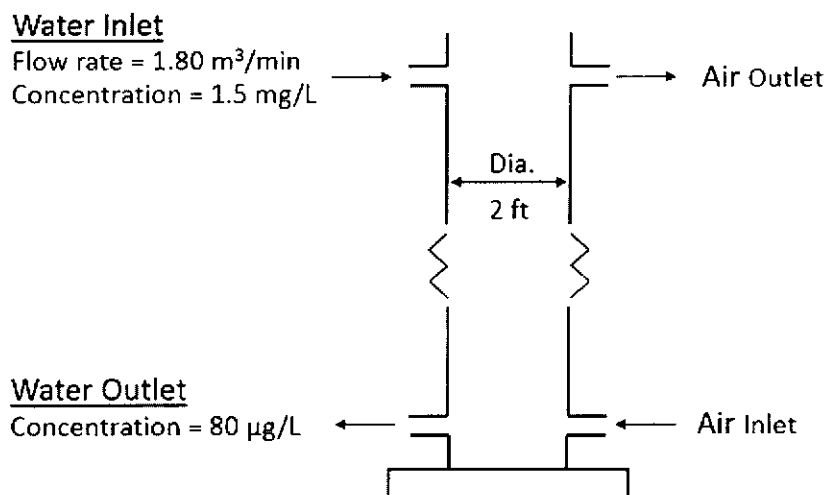
- (c) Draw a typical groundwater monitoring well with its components. (08)
- (d) Three wells are set up at the East, West and South sides to monitor the ground water of a specific site. From the following figure, determine the equipotential line and hydraulic gradient of this setup. You must show the direction of flow among the wells. (15)



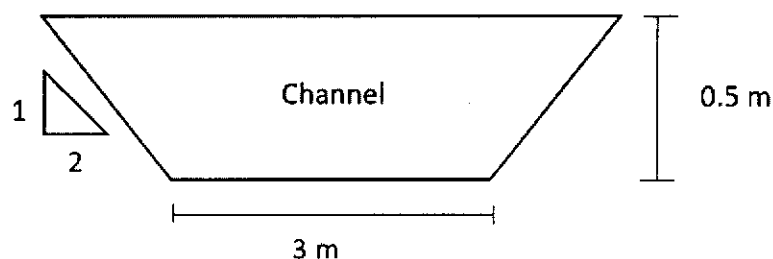
2. (a) Define air stripping process in water treatment. (05)
- (b) Describe the following layout of distribution pipe network with necessary sketches: (12)
 - i) Ring or circular system ii) Tree end system iii) Grid or interlaced system
- (c) Suppose you are working in the city corporation authority, where an industry of your area is spilling hazardous substances into an aquifer. There is a river near to that aquifer. From the following figure determine how many days do you have to save the river resources from that spill? (18)
 - Hydraulic conductivity of aquifer = 217 m/day
 - Porosity = 0.30
 - Retardation factor = 1.5



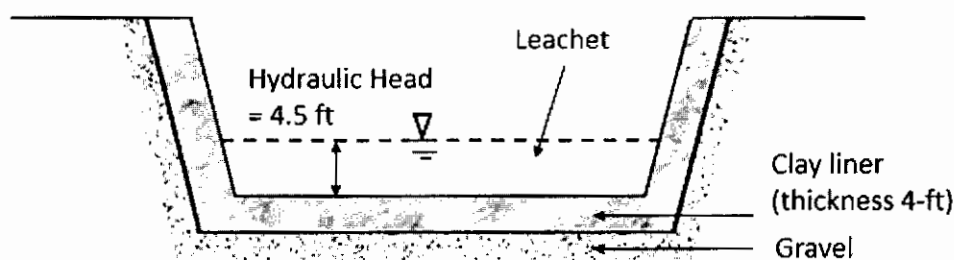
3. (a) What are the conditions to use combined sewerage systems? (05)
- (b) Design an air stripping water treatment column to remove 1,1,1-Trichloroethane (15)
- from the water, using the following information and figure:
- Use 2-in Intalox saddles for packing (area to volume ratio is 118 m^{-1}).
 - Column diameter = 2 ft
 - Liquid mass transfer coefficient = 16.5 cm/hr .
 - Air-to-water ratio = 65
 - Henry's constant (dimensionless) = 0.464
 - Also consider a safety factor (SF) = 20%.



- (c) A trapezoidal channel has concrete-steel troweled (0.014) material, and slope of hydraulic gradient = 0.002, and side slope of trapezoid is 1 vertical to 2 horizontals. Calculate the flow rate using the following figure: (15)



4. (a) Describe the following components of a sanitary landfill: (12)
- i) Active face ii) Alternative daily cover iii) Final covers
- (b) Draw and show cross section of a typical waste landfill with EPA recommended cover design. (13)
- (c) If the break-through time for leachet to penetrate a 4-ft thick clay liner is 10 years estimate the coefficient of permeability (cm/s). Given that the effective porosity is 0.30. Use the following figure for other information. (10)



Section- B

5. (a) Infrastructure includes physical facilities and their services. how so? Briefly explain. (10)
- (b) Narrate the ways of overcoming infrastructure-related crises with relevant examples. (10)
- (c) Discuss about the supply and demand side mechanism that connects infrastructure to economic activities. (15)
6. (a) 'The values of social and physical infrastructure are interconnected' - Discuss with an example. (05)
- (b) 'Climate change drive infrastructure investment decisions' - Explain the statement with example. (10)
- (c) How to demand side constants make it difficult to recover the costs of new infrastructure investment? (10)
- (d) Mention some points which are necessary to consider for setting the price of infrastructure. (10)
7. (a) Suppose, you are going to choose a suitable site for the development of a major infrastructure. Based on cost, transportation, labor availability and quality of life, there are three available alternative sites. The sites are: Location A, Location B and Location C. Comparison of the mentioned criteria and the ranking alternatives are given in the following tables (35)

i) Criteria matrix:

Criteria	Cost	Transportation	Labor availability	Quality of life
Cost	1	1/2	3	2
Transportation	2	1	5	3
Labor availability	1/3	1/3	1	1/2
Quality of life	1/2	1/3	2	1

ii) Matrix for Cost:

Cost	Location A	Location B	Location C
Location A	1	2	5
Location B	1/2	1	3
Location C	1/5	1/3	1

iii) Matrix for Transportation:

Transportation	Location A	Location B	Location C
Location A	1	1/3	2
Location B	3	1	5
Location C	1/2	1/5	1

iv) Matrix for Labor availability

Labor availability	Location A	Location B	Location C
Location A	1	1/2	3
Location B	2	1	4
Location C	1/3	1/4	1

v) Matrix for Quality of life

Quality of life	Location A	Location B	Location C
Location A	1	3	2
Location B	1/3	1	1/2
Location C	1/2	2	1

Select the best alternative for the development of a major infrastructure using Analytical Hierarchy Process (AHP).

8. (a) Define the following terms associated with infrastructure (Any Five): (35)
- i) Earmarked account
 - ii) Debt and equity
 - iii) Grants
 - iv) Spillover benefit
 - v) Financing
 - vi) Funding
 - vii) Crowding in